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CPS 5745


Class Activity and Reflection:

During each classmate's presentation, take brief notes recording the project's title, main scientific/educational question, one new insight you learned, how an interaction changes the visualizations, and one specific strength and suggested improvement.

Contribute at least one thoughtful question or constructive comment during the presentation session, referring to a particular visualization, finding, or design choice.

After all presentations, submit your notes and a 150–200-word reflection summary comparing at least two classmates' projects: what stood out the most? What surprised you or challenged your understanding? What specific idea or improvement will you apply to your own project? Support your comments and reflection with examples from the live demonstrations.

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**1. Danny Nunoz — *How Fast Does a Used Ford Lose Its Value?***

- **Question:** How do age, mileage, and car features relate to used Ford prices?
- **Insight:** The data compares different cars at one moment; it does not follow the same car as it loses value.
- **Interaction:** Changing the car's age or mileage updates its estimated price and comparison charts.
- **Strength:** The project shows prediction errors using cars held out from training.
- **Improvement:** Make the assumed listing year, 2020, more noticeable beside the age controls.
- **Question to ask:** "Could differences between model years explain some of what looks like depreciation?"

**2. Noor Balouch — *Powering the Future***

- **Question:** How have renewable electricity generation and its share changed across countries, and how do their energy mixes differ?
- **Insight:** Renewable electricity share (%) and the amount generated (TWh) answer different questions.
- **Interaction:** Changing the map year updates country shares; choosing countries changes the comparison lines.
- **Strength:** Missing observations remain gaps instead of being presented as zero.
- **Improvement:** Add a short takeaway identifying the largest change among the selected countries.

- **Question to ask:** "Can a country increase renewable generation while its renewable share decreases?"

**3. Mitchell Toran — *Student Lifestyle and Stress***

- **Question:** How do students' daily habits relate to GPA and stress?
- **Insight:** High stress accounts for about 51% of the 2,000 students in this dataset.
- **Interaction:** The sleep view's stress filter lets users examine sleep and GPA within a selected stress group.
- **Strength:** Color shows stress alongside sleep and GPA in the same scatterplot.
- **Improvement:** Display the number of students remaining after each filter.
- **Question to ask:** "Does the sleep–GPA pattern look different within each stress group?"

**4. Jonayed Hossain Chowdhury — *Counting Is Not Risk: Rain, Darkness and the Missing Denominator***

- **Question:** Why do more reported crashes in dry daylight conditions not necessarily mean those conditions are riskier?
- **Insight:** Crash counts need a denominator, such as miles driven, before they can describe risk.
- **Interaction:** Changing the assumed share of driving in rain changes the relative crash-rate comparison.
- **Strength:** The project clearly separates reported data, physical simulations, and user assumptions.
- **Improvement:** Place a simple worked example beside the exposure slider.
- **Question to ask:** "How sensitive is the risk ranking to uncertainty in the miles-driven estimate?"

**5. Karanpreet Singh — *From Pixels to Weather***

- **Question:** How can labeled photographs become data for weather-image classification?
- **Insight:** "Sunrise" describes a scene or time of day and can overlap visually with cloudy or sunny conditions.
- **Interaction:** The weather-class filter changes the image gallery; Inspect opens a photograph and its properties.
- **Strength:** The project separates training examples from validation and test views.

- **Improvement:** Add a comparison of visually similar photographs with different labels.
- **Question to ask:** "How does overlap between sunrise and other labels affect classification mistakes?"

**6. Jean Stephane Yapo — *Meshline: Network Simulation***

- **Question:** How do routing algorithms, traffic, and link failures affect packet movement?
- **Insight:** A route can minimize total latency while passing through several intermediate routers.
- **Interaction:** Cutting a link changes the available network; the failover view shows alternate routes.
- **Strength:** The Dijkstra view includes a distance table and step-by-step explanation.
- **Improvement:** Clearly distinguish illustrative dashboard metrics from values calculated by the simulation.
- **Question to ask:** "Under what conditions would Dijkstra and Bellman–Ford choose different routes?"

**7. Cristopher Rincon Padilla — *A Jam with No Cause / Phantom Traffic Jam***

- **Question:** Can traffic jams form without an accident or obstruction, and can an automated car reduce them?
- **Insight:** A stop-and-go wave can travel backward while the cars themselves move forward.
- **Interaction:** Changing the number of automated cars in the comparison changes the simulated traffic behavior and speed-variation chart.
- **Strength:** The two-ring comparison uses the same random driver noise, helping isolate the automated-car difference.
- **Improvement:** Add a plain-language definition of "headway" beside its slider.
- **Question to ask:** "Why does the simulation smooth traffic more completely than the real experiment?"

**8. David Arosemena — *Catching a Quantum Spy***

- **Question:** Can someone intercept a quantum key without leaving detectable errors?

- **Insight:** Ordinary channel noise also creates errors, making evidence of interception harder to interpret.
- **Interaction:** Increasing channel noise raises the error-rate curve; increasing the number of checked bits raises the modeled detection probability.
- **Strength:** The project compares simulation results with theoretical predictions and shows uncertainty.
- **Improvement:** Keep a short definition of “sifted key” beside the error-rate charts.
- **Question to ask:** “How does normal channel noise affect confidence that an error came from Eve?”